

MTL106: Probability and Stochastic Processes
Lecture 13
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Conditional Probability Mass Function: Let (X, Y) be a discrete random variable. If $P(Y = y_j) > 0$, the function

$$P\{X = x_i | Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}} \quad (1)$$

for fixed j is known as the *conditional probability mass function* of X , given $Y = y_j$. Similarly, If $P(X = x_i) > 0$, the function

$$P\{Y = y_j | X = x_i\} = \frac{P\{X = x_i, Y = y_j\}}{P\{X = x_i\}} \quad (2)$$

for fixed i is known as the *conditional probability mass function* of Y , given $X = x_i$.

For the joint PMF of Example 2 (in Lecture 12), we have for $Y = 1$,

$$P\{X = 0 | Y = 1\} = 3/72, \quad P\{X = 1 | Y = 1\} = 5/72, \quad P\{X = 2 | Y = 1\} = 7/72.$$

Conditional Probability Density Function: Let f be the PDF of a continuous random variable (X, Y) , and let f_2 be the marginal PDF of Y . At every point (x, y) at which f is continuous and $f_2(y) > 0$ and is continuous, the *conditional probability density function* of X , given Y , exists and is expressed by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_2(y)}. \quad (3)$$

Similarly, let f_1 be the marginal PDF of X . At every point (x, y) at which f is continuous and $f_1(x) > 0$ and is continuous, the *conditional probability density function* of Y , given X , exists and is expressed by

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_1(x)}. \quad (4)$$

For the joint PDF of Example 3 (in Lecture 12), the conditional PDF of X given Y is given by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_2(y)} = \frac{2}{2y} = \frac{1}{y}, \quad 0 < x < y.$$

Conditional Cumulative Distribution Function: Let (X, Y) be a random variable with CDF F . Then the *conditional cumulative distribution function* of X given Y is defined by

$$F_{X|Y}(x|y) = \begin{cases} \sum_{x_i \leq x} P\{X = x_i | Y = y_j\}, & \text{if } (X, Y) \text{ is discrete,} \\ \int_{-\infty}^x f_{X|Y}(t|y) dt, & \text{if } (X, Y) \text{ is continuous.} \end{cases} \quad (5)$$

A similar definition can be given for the conditional CDF of Y given X .

Independent Random Variables: We say that two random variables X and Y are independent if and only if

$$F(x, y) = F_1(x)F_2(y) \quad \forall (x, y) \in \mathbb{R}_2. \quad (6)$$

Lemma 1: If the random variables X and Y are independent and $a < c$, $b < d$ are real numbers, then

$$P\{a < X \leq c, b < Y \leq d\} = P\{a < X \leq c\}P\{b < Y \leq d\}. \quad (7)$$

Theorem 1: (a) A necessary and sufficient condition for the discrete random variables X and Y to be independent is that

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\}P\{Y = y_j\} \quad (8)$$

for all pairs (x_i, y_j) .

(b) Two continuous random variables X and Y are independent if and only if

$$f(x, y) = f_1(x)f_2(y) \quad \forall (x, y) \in \mathbb{R}_2, \quad (9)$$

where f, f_1, f_2 , respectively, are the joint and marginal densities of X and Y , and f is everywhere continuous.

Proof: (a) Let X, Y be independent. Then from Lemma 1, letting $a \rightarrow c$ and $b \rightarrow d$, we get

$$P\{X = c, Y = d\} = P\{X = c\}P\{Y = d\}.$$

Conversely

$$\begin{aligned} F(x, y) &= \sum_{x_i \leq x, y_j \leq y} P\{X = x_i, Y = y_j\}. \\ \Rightarrow F(x, y) &= \sum_{x_i \leq x} \left[\sum_{y_j \leq y} P\{Y = y_j\} \right] P\{X = x_i\} = F(x)F(y). \end{aligned}$$

(b) The proof of part (b) is trivial.

Remark 1: Let X and Y be independent random variables, then

$$F_{X|Y}(x|y) = F_X(x) \quad \forall x, \quad \text{and} \quad F_{Y|X}(y|x) = F_Y(y) \quad \forall y.$$

Remark 2: The random variables X and Y are independent if and only if

$$P\{X \in A_1, Y \in A_2\} = P\{X \in A_1\}P\{Y \in A_2\} \quad (10)$$

for all Borel sets A_1 on the x -axis and A_2 on the y -axis.

Theorem 2: Let X and Y be independent random variables and f and g be Borel-measurable functions. Then $f(X)$ and $g(Y)$ are also independent.

Proof: We have

$$\begin{aligned} P\{f(X) \leq x, g(Y) \leq y\} &= P\{X \in f^{-1}(-\infty, x], Y \in g^{-1}(-\infty, y]\} \\ &= P\{X \in f^{-1}(-\infty, x]\}P\{Y \in g^{-1}(-\infty, y]\} \\ &= P\{f(X) \leq x\}P\{g(Y) \leq y\}. \end{aligned}$$

Example 1: Let X and Y be jointly distributed with PDF

$$f(x, y) = \begin{cases} \frac{1+xy}{4}, & |x| < 1, |y| < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then the marginal densities of X and Y are $f_1(x) = 1/2$, $|x| < 1$ and $f_2(y) = 1/2$, $|y| < 1$ respectively. Hence X and Y are not independent, since $f(x, y) \neq f_1(x)f_2(y)$. However, the random variables X^2 and Y^2 are independent, since one can easily prove, $P\{X^2 \leq u, Y^2 \leq v\} = P\{X^2 \leq u\}P\{Y^2 \leq v\}$.

Independence of n Random Variables: A collection of jointly distributed random variables $X_1, X_2, \dots, X_n$ is said to be mutually or completely independent if and only if

$$F(x_1, x_2, \dots, x_n) = F_1(x_1)F_2(x_2)\dots F_n(x_n) \quad \forall (x_1, x_2, \dots, x_n) \in \mathbb{R}_n, \quad (11)$$

where F is the joint CDF of $(X_1, X_2, \dots, X_n)$, and $F_i (i = 1, 2, \dots, n)$ is the marginal CDF of X_i .

The random variables $X_1, X_2, \dots, X_n$ are said to be pairwise independent if and only if every pair of them are independent.

Remark 3: If $X_1, X_2, \dots, X_n$ are independent, every subcollection $X_{i_1}, X_{i_2}, \dots, X_{i_k}$ of $X_1, X_2, \dots, X_n$ is also independent.

Remark 4: Pairwise independence may not imply mutual independence.

Example 2: Let X, Y, Z have the joint PMF defined by

$$P\{X = x, Y = y, Z = z\} = \begin{cases} \frac{3}{16}, & \text{if } (x, y, z) \in \{(0, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)\}, \\ \frac{1}{16}, & \text{if } (x, y, z) \in \{(0, 0, 1), (0, 1, 0), (1, 0, 0), (1, 1, 1)\}. \end{cases}$$

We have

$$P\{X = x, Y = y\} = \frac{1}{4}, \quad (x, y) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

$$P\{Y = y, Z = z\} = \frac{1}{4}, \quad (y, z) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

$$P\{X = x, Z = z\} = \frac{1}{4}, \quad (x, z) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

$$P\{X = x\} = \frac{1}{2}, \quad x = 0, 1,$$

$$P\{Y = y\} = \frac{1}{2}, \quad y = 0, 1,$$

$$P\{Z = z\} = \frac{1}{2}, \quad z = 0, 1.$$

Clearly X and Y , Y and Z , and X and Z are pairwise independent. However, X, Y, Z are not independent.

Remark 5: A sequence $\{X_n\}$ of random variables is said to be independent if for every $n = 2, 3, 4, \dots$ the random variables $X_1, X_2, \dots, X_n$ are independent.

Remark 6: The random variables X and Y are identically distributed if X and Y have the same CDF, i.e., $F_X(x) = F_Y(x) \quad \forall \quad x \in \mathbb{R}$, where F_X and F_Y are the CDFs of X and Y , respectively. Note that X and Y are identically distributed if and only if they have the same distribution, It does not follow that $X = Y$ with probability 1. If $P\{X = Y\} = 1$, we say that X and Y are equivalent random variables. So basically we want to say that X and Y are identically distributed if and only if $P\{X \in A\} = P\{Y \in A\} \quad \forall \quad A \in \mathcal{B}$. Nothing is said about the equality of events $\{X \in A\}$ and $\{Y \in A\}$.

Remark 7: We say that $\{X_n\}$ is a sequence of independent, identically distributed (iid) random variables with common law $\mathcal{L}(X)$ if $\{X_n\}$ is an independent sequence of random variables and the distribution of X_n ($n = 1, 2, \dots$) is same as that of X .